

Networked predictive control for linear systems with unknown communication delay

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Abstract—The problem of controller design for linear systems with unknown communication delay is investigated in this paper. A control scheme is proposed to deal with the unknown communication delay on the basis of networked predictive control method. By considering an important property of communication delay, the closed-loop system is modeled as a switched system under constraint switching. By using switched Lyapunov function approach, a sufficient stability condition is derived. How to choose an important parameter in the control law is also considered. Finally, two numerical examples are given to illustrate the effectiveness of the proposed method.

Keywords—Networked control system, Networked predictive control, Unknown delay, Switched system

I. INTRODUCTION

Recently, networked control systems (NCSs) have been a hot research topic and rich papers about the topic have been published. Along with advantages such as low power requirements and high flexibility, some disadvantages such as communication delay, data drops, are also introduced in NCSs [1], [2].

To deal with problems caused by network introduction, many methodologies have been proposed. For example, in [3], [4], the Lyapunov-Krasovskii method widely used in time-delay systems was used to derive delay-dependent stability conditions for NCSs. NCSs can be modeled as time-delay systems where the time-delay is non-differentiable. In [5], two Markov chains were used to model the sensor-to-controller and the controller-to-actuator communication delay and the closed-loop system was model as a jump system with two modes and a state feedback controller design method was proposed. Results in [5] were further extended in [6] under some relaxed assumptions. In [7], a new model for NCSs was established taking network-induced delay, time-varying sampling intervals and data dropouts into account and a state feedback controller was designed based on this model. In [8], a convex overapproximation method was proposed to deal with time-varying transmission intervals and transmission constraints in NCSs. Necessary and sufficient conditions on the problem of minimum data rate for mean square stabilization of NCSs have been investigated in [9], [10] where the data dropouts were model as an independent and identically distributed process and time-homogeneous Markov process respectively. Other methodologies refer to some survey papers [11], [12], [13] and references therein.

In most existing publications, the communication delay that a packet suffers is often assumed to be known by the controller. However, it is not always the case in practical systems. The communication delay is not easy to measure since it needs that all the components in the system are all synchronized. In practical MCSs, synchronization of all the components may be an expensive or tough task. Compared with rich results concerning with NCSs with known communication delays, results about NCSs with unknown communication delays are much little. By employing Taylor Series Expansion, the controller design problem for NCSs with unknown time-varying communication delays was transformed into a problem of stabilization of uncertain systems with polytopic uncertainties [14]. The unknown communication delay was transformed into norm-bounded uncertainties of the system's parameters, robust delay-dependent stability conditions were obtained and stabilization controller design problem was investigated [15].

In addition, most existing literature aim to design a controller which is robust to communication delays but not consider how to compensate for network-induced delay actively. Recently, a so-called networked predictive control method has been proposed to compensate for network-induced delay [16], [17], [18], [19], [20], [21], [22]. Both simulations and experimental results show that this method is quite effective and can obtain a similar performance as local control. However, the network-induced delay is assumed to be known in networked predictive control method. Based on this observation, a natural question is as follows: How to design a networked predictive controller when the network-induced delay is unknown?

In this paper, an answer to the above question is given. A controller design scheme which is robust to the unknown communication delay is proposed based on the networked predictive control method. By taking an important property of communication delay into account, the closed-loop system is transformed into a switched system under constraint switching. A sufficient stability condition is derived using switched Lyapunov function approach and presented in terms of linear matrices inequality. Finally, two numerical examples are given to confirm the validity and effectiveness of the proposed method.

II. FORMULATION

A discrete-time linear system is described by

$$x(k+1) = Ax(k) + Bu(k) \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the state vector; $u(k) \in \mathbb{R}^m$ is the control input; $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$ are constant system matrices in companion form.

For the sake of simplicity, but without loss of generality, networked control systems with random delay in the sensor-to-controller channel are considered in this paper. The network communication delay $\tau(k)$ is assumed to be unknown but vary within an known interval $\tau_1 \leq \tau(k) \leq \tau_2$.

Following the networked predictive control scheme [16], [17], [18], [19], [20], [21], [22], if the network-induced delay is known, based on the received state $x(k-\tau(k))$ at the controller node, the current state can be predicted by the following recursive iteration.

$$\begin{aligned} x(k-\tau(k)+1) &= Ax(k-\tau(k)) + Bu(k-\tau(k)) \\ x(k-\tau(k)+2) &= Ax(k-\tau(k)+1) \\ &\quad + Bu(k-\tau(k)+1) \\ &= A^2x(k-\tau(k)) + ABu(k-\tau(k)) \\ &\quad + Bu(k-\tau(k)+1) \\ &\vdots \\ x(k) &= A^{\tau(k)}x(k-\tau(k)) \\ &\quad + \sum_{i=1}^{\tau(k)} A^{i-1}Bu(k-i) \end{aligned} \quad (2)$$

From the predicted state $x(k)$, a state feedback controller can be designed as

$$\begin{aligned} u(k) &= Kx(k) \\ &= KA^{\tau(k)}x(k-\tau(k)) \\ &\quad + K \sum_{i=1}^{\tau(k)} A^{i-1}Bu(k-i) \end{aligned} \quad (3)$$

where K is the controller gain matrix.

When the communication delay $\tau(k)$ is unknown, the above networked predictive control scheme is unavailable. Therefore, the following state feedback controller is proposed to deal with the unknown communication delay.

$$u(k) = KA^h x(k-\tau(k)) + K \sum_{i=1}^h A^{i-1} Bu(k-i) \quad (4)$$

where h is the prediction step and satisfies $\tau_1 \leq h \leq \tau_2$. How to choose an appropriate h will also be considered in this paper.

Similar to [16], [17], [18], [19], [20], [21], [22], the controller gain matrix K can be designed without considering the network introduction and can be determined by some existing methods such as LQ and pole assignment. It should be noted that the controller gain matrix K can also be determined by solving the stability conditions in Theorem 1 in the following part.

III. STABILITY ANALYSIS

In this section, stability of the closed-loop system is analyzed and a sufficient stability condition is obtained using switched Lyapunov function approach.

Substituting (4) into (1) yields

$$\begin{aligned} x(k+1) &= Ax(k) + BKA^h x(k-\tau(k)) \\ &\quad + BK \sum_{i=1}^h A^{i-1} Bu(k-i) \end{aligned} \quad (5)$$

Clearly,

$$x(k) = A^h x(k-h) + \sum_{i=1}^h A^{i-1} Bu(k-i) \quad (6)$$

Therefore,

$$\sum_{i=1}^h A^{i-1} Bu(k-i) = x(k) - A^h x(k-h) \quad (7)$$

From (5) and (7), one can obtain that

$$\begin{aligned} x(k+1) &= (A + BK)x(k) + BKA^h x(k-\tau(k)) \\ &\quad - BKA^h x(k-h) \end{aligned} \quad (8)$$

The closed-loop system (8) can be seen as a system with two time-delays. One is constant and the other is time-varying. Therefore, we can use Lyapunov-Krasovskii functional method [23], [24] to obtain some stability conditions. In NCSs, it is natural that the controller always use the most latest data. If the delayed system state $x(k-\tau(k))$ is available at step k but there is no new information arriving at step $k+1$, then $x(k-\tau(k))$ is at least available for controller design. Therefore, the network-induced delay increases at most by 1 each step, that is, $\tau(k+1) \leq \tau(k) + 1$. [23], [24] have only considered the interval delay case but ignore this important property of network-induced delay $\tau(k)$. Taking this property into account, the closed-loop system (8) can be transformed into a switched system under constraint switching.

Define a new vector $z(k) = \begin{bmatrix} x(k) \\ x(k-1) \\ \vdots \\ x(k-\tau_2+1) \\ x(k-\tau_2) \end{bmatrix}$ and

system (8) can be rewritten as the following switched system

$$z(k+1) = \Upsilon_{\sigma(k),h} z(k) \quad (9)$$

where $\sigma(k) \in \mathcal{I}$ is a piecewise constant function and denotes the active mode. $\mathcal{I} = \{1, 2, \dots, \bar{\tau}\}$, $\bar{\tau} = \tau_2 - \tau_1 + 1$,

$$\begin{aligned} \Upsilon_{i,h} &= \begin{bmatrix} 0 & \overbrace{0 \cdots 0}^{i+\tau_1-2} & A^h BK & 0 & \cdots & 0 \\ & & \mathbf{0}_{\tau_2 n \times \tau_2 n} & & & 0 \\ & & & & & \vdots \\ & & & & & 0 \end{bmatrix} \\ &+ \begin{bmatrix} A & \overbrace{0 \cdots 0}^{h-1} & -A^h BK & 0 & \cdots & 0 \\ & & \mathbf{I}_{\tau_2 n \times \tau_2 n} & & & \vdots \\ & & & & & 0 \end{bmatrix}, \quad i \in \mathcal{I} \end{aligned}$$

Since the network-induced delay satisfies $\tau(k+1) \leq \tau(k) + 1$, it is easy to see that $\sigma(k+1) \leq \sigma(k) + 1$. Therefore, the switched system (9) is under constraint switching but not arbitrary switching.

Using switched Lyapunov function approach [25], a sufficient stability condition for system (9) is derived.

Theorem 1. For given controller gain matrix K , if there exist matrices $P_i > 0$ and any matrices G_i, F_i such that

$$\begin{bmatrix} F_i \Upsilon_{i,h} + \Upsilon_{i,h}^T F_i^T - P_i & * \\ G_i \Upsilon_{i,h} - F_i^T & P_j - G_i - G_i^T \end{bmatrix} < 0 \quad (10)$$

for all $(i, j) \in \mathcal{I} \times \mathcal{I}$ and $j \leq i + 1$, then system (9) is asymptotically stable

Proof: Follow the method in [26] and choose a switched Lyapunov function

$$\begin{aligned} V(k, z_k) &= z(k)^T P(\alpha(k)) z(k) \\ &= z^T(k) \left(\sum_{i=1}^{\bar{\tau}} \alpha_i(k) P_i \right) z(k) \end{aligned} \quad (11)$$

where $\alpha(k)$ is the indicator function as defined in [26].

One can obtain that

$$\begin{aligned} \Delta V &= V(k+1, z_{k+1}) - V(k, z_k) \\ &= z^T(k+1) P(\alpha(k+1)) z(k+1) \\ &\quad - z(k)^T P(\alpha(k)) z(k) \\ &= z^T(k+1) P_j z(k+1) - z^T(k) P_i z(k) \end{aligned} \quad (12)$$

It is easy to see that the following equation holds for all $(i, j) \in \mathcal{I} \times \mathcal{I}$ and $j \leq i + 1$

$$0 = 2 [z^T(k+1) G_i + z^T(k) F_i] [-z(k+1) + \Upsilon_{i,h} z(k)] \quad (13)$$

Adding both sides of (13) to both sides of (12) yields

$$\begin{aligned} \Delta V &= z^T(k+1) (P_j - G_i - G_i^T) z(k+1) \\ &\quad + z^T(k) (F_i \Upsilon_{i,h} + \Upsilon_{i,h}^T F_i^T - P_i) z(k) \\ &\quad + 2z^T(k+1) (G_i \Upsilon_{i,h} - F_i^T) z(k) \end{aligned} \quad (14)$$

So if (10) holds, then $\Delta V < 0$ which means that system (9) is asymptotically stable. ■

From Corollary 2 in [26], we can see that if the system (9) is asymptotically stable, it is also exponentially stable. A decay rate can be estimated by the following theorem.

Theorem 2. For given controller gain matrix K , system (9) is exponentially stable and the decay rate is $\lambda = \sqrt{1-v}$, if there exist matrices $P_i > 0$, any matrices G_i, F_i and scalars $v > 0, \eta > 0$, such that

$$\begin{bmatrix} F_i \Upsilon_{i,h} + \Upsilon_{i,h}^T F_i^T - P_i + vI & * \\ G_i \Upsilon_{i,h} - F_i^T & P_j - G_i - G_i^T \end{bmatrix} < 0 \quad (15)$$

$$\eta I < P_i < I \quad (16)$$

for all $(i, j) \in \mathcal{I} \times \mathcal{I}$ and $j \leq i + 1$.

Proof: The proof can be completed following the similar line as Theorem 3 in [26] and is omitted here. ■

IV. DESIGN OF THE PREDICTION STEP h

From the control law (4), it can be seen that h is an important parameter. How to choose an appropriate h will be considered in this section. The basic idea is that an appropriate h should make the decay rate of the closed-loop system as large as possible.

Theorem 3. For given controller gain matrices K , an appropriate h for system (8) is given by $h = \arg \max \{v_k^{opt}, k \in \mathcal{I}\} + \tau_1 - 1$, where v_k^{opt} is the optimal value of the following optimization problem

Maximize v_k

Subject to

$$\begin{bmatrix} F_i \Upsilon_{i,\hat{k}} + \Upsilon_{i,\hat{k}}^T F_i^T - P_i + v_k I & * \\ G_i \Upsilon_{i,\hat{k}} - F_i^T & P_j - G_i - G_i^T \end{bmatrix} < 0 \quad (17)$$

$$\eta I < P_i < I \quad (18)$$

for all $(i, j) \in \mathcal{I} \times \mathcal{I}$ and $j \leq i + 1$ with $\Upsilon_{i,\hat{k}} = \Upsilon_{i,k+\tau_1-1}$ being of the following form

$$\begin{aligned} \Upsilon_{i,\hat{k}} &= \begin{bmatrix} 0 & \overbrace{0 \cdots 0}^{i+\tau_1-2} & A^{\hat{k}} B K & 0 & \cdots & 0 \\ & & \mathbf{0}_{\tau_2 n \times \tau_2 n} & & & 0 \\ & & & & & \vdots \\ & & & & & 0 \end{bmatrix} \\ &+ \begin{bmatrix} A & \overbrace{0 \cdots 0}^{\hat{k}-1} & -A^{\hat{k}} B K & 0 & \cdots & 0 \\ & & \mathbf{I}_{\tau_2 n \times \tau_2 n} & & & \vdots \\ & & & & & 0 \end{bmatrix}, \quad i \in \mathcal{I} \end{aligned}$$

V. NUMERICAL EXAMPLES

In this section, two numerical examples are given to show the effectiveness of the method developed in this paper.

Example 1. Consider the following cart-pendulum system [23] whose simplified and discretized model is

$$\begin{aligned} A &= \begin{bmatrix} 1.0078 & 0.0301 \\ 0.5202 & 1.0078 \end{bmatrix}, \\ B &= \begin{bmatrix} -0.0001 \\ -0.0053 \end{bmatrix}, \quad C = [1 \quad 0] \end{aligned}$$

The eigenvalues of matrix A are 1.1329 and 0.8827, which means that the open-loop system is unstable. The feedback control gain matrix K is designed by pole assignment method to ensure the closed-loop system without network-induced delay is stable. The desired poles of the feedback controller is $[0.8, 0.9]$ and K is obtained as follows

$$K = [220.5080 \quad 55.3866]$$

The network-induced delay is bounded by $1 \leq \tau(k) \leq 5$ as shown in Fig.1.

Using Theorem 3, we can obtain an appropriate $h = 3$. Let $x(0) = [10 \quad 0]^T$, simulation results are given in Fig.2. We can see that the closed-loop system is asymptotically stable.

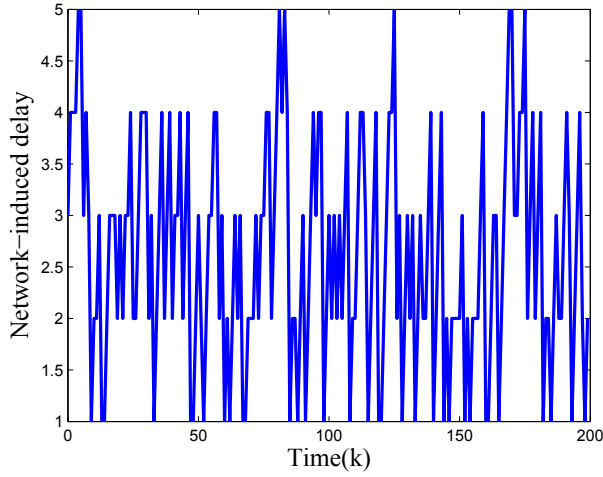


Fig. 1. Network-induced delay of example 1

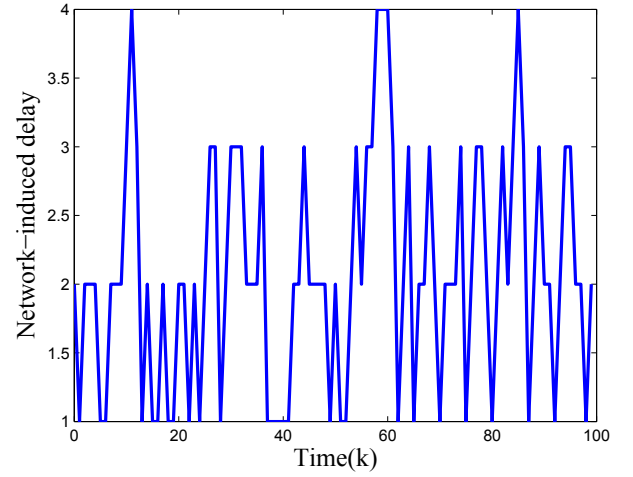


Fig. 3. Network-induced delay of example 2

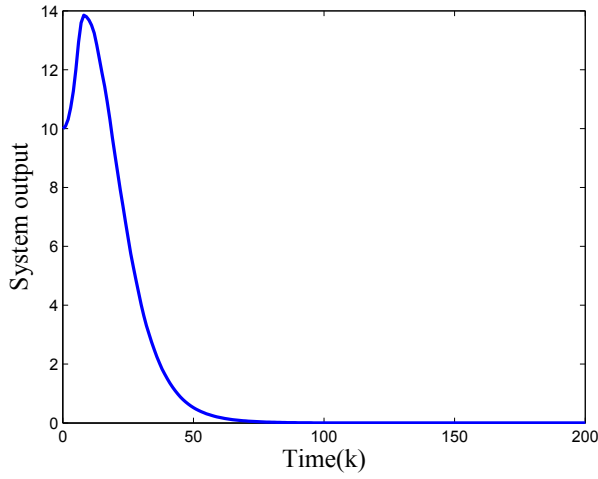


Fig. 2. System output of example 1

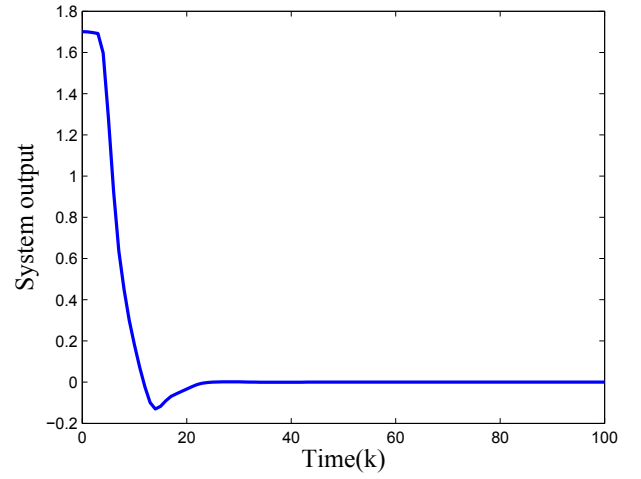


Fig. 4. System output of example 2

Example 2. Consider the servo control system in [23] whose discrete-time model is described by

$$A = \begin{bmatrix} 1.120 & 0.213 & -0.335 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

$$B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad C = [0.0541 \quad 0.115 \quad 0.0001]$$

Let the desired poles of the feedback controller is $[-0.3, 0.8 + 0.3i, 0.8 - 0.3i]$ and controller gain matrix K is

$$K = [0.1800 \quad -0.4630 \quad 0.1160]$$

The network-induced delay is bounded by $1 \leq \tau(k) \leq 4$ as shown in Fig.3.

Using Theorem 3, an appropriate h is can be obtained as $h = 3$. Let $x(0) = [10 \ 10 \ 10]^T$, simulation results are given in Fig.4. We can see that the closed-loop system is asymptotically stable.

VI. CONCLUSIONS

The problem of controller design for networked control systems with unknown time-varying communication delay has been investigated in this paper. A modified networked predictive control scheme has been proposed to deal with the unknown delay. The closed-loop system has been converted to a discrete-time switched system. Using switched Lyapunov functional method, sufficient stability conditions have been obtained. How to choose a key parameter in the control law has been also considered. Finally, effectiveness of the proposed method has been illustrated by two numerical examples.

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